

# **CourseCraft AI**

**AI-Integrated Intelligent Learning Management System**

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## Certificate of Examination

This is to certify that the project report entitled '**CourseCraft AI**' submitted by **Subhankar Patra (2302080033)** and **Shibani Taria (2302080001)** of Information Technology & Engineering Branch, Veer Surendra Sai University of Technology, Odisha has been examined by me. I am satisfied with the quality and correctness of the work.

Supervisor



## Supervisor's Certificate

This is to certify that the Project-I work entitled “**CourseCraft AI**” is being submitted by **Subhankar Patra(2302080033)** and **Shibani Taria(2302080001)** to the Department of Computer Science & Engineering, Veer Surendra Sai University of Technology, Burla, in partial fulfillment of the requirement for the degree of Bachelor of Technology in Information Technology & Engineering during the academic year 2025-2026. It is an original report carried out by them under my supervision.

**Supervisor**

# **Acknowledgement**

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Date: 7th of April , 2026  
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# Abstract

In today's fast-moving tech world, many beginners find it hard to pick a specific field and follow a clear learning path. Computer science has many different areas, which can feel overwhelming for new learners. While there is a lot of free educational content on platforms like YouTube, these resources are often unorganized and lack a proper step-by-step sequence. Students often spend too much time searching for the right videos instead of actually studying. Also, long video lectures can be inefficient, and many learners prefer clear, well-structured text over long videos.

To solve this, the "CourseCraft AI" system collects free content and organizes it into structured courses. It uses Natural Language Processing (NLP) and Large Language Models (LLMs) to turn video lectures into easy-to-read study notes. The system takes YouTube transcripts, breaks them into smaller parts to fit model limits, and organizes them into a Domain -> Course -> Module -> Lesson structure. It also includes a Course-Aware Retrieval-Augmented Generation (RAG) chatbot to answer student questions using only the information from the generated course.

**Keywords:** Artificial Intelligence, Natural Language Processing, Large Language Models, Retrieval-Augmented Generation, Learning Management System, Text Summarization.

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## **Abbreviations:**

**AI:** Artificial Intelligence

**API:** Application Programming Interface

**CDN:** Content Delivery Network

**CRUD:** Create, Read, Update, Delete

**DBMS:** Database Management System

**JWT:** JSON Web Token

**LLM:** Large Language Model

**LMS:** Learning Management System

**NLP:** Natural Language Processing

**ORM:** Object-Relational Mapping

**RAG:** Retrieval-Augmented Generation

**UI/UX:** User Interface / User Experience

# 1. Introduction

In today's fast-moving digital landscape, the overwhelming abundance of unstructured data often makes it difficult for beginners to establish a clear and effective learning path. While platforms like YouTube offer an unprecedented volume of high-quality instructional content, these resources frequently lack the pedagogical scaffolding necessary to transform raw media into organized knowledge. Consequently, learners experience significant cognitive overload and spend far too much time navigating disjointed video lectures, leading to major temporal inefficiencies.

At present, there is a distinct lack of intelligent systems capable of systematically organizing scattered content into cohesive courses, converting long video lectures into concise study notes, and providing a guided, well-structured learning experience for novice students.

The proposed "**CourseCraft AI**" system serves as a technological solution designed to aggregate unstructured video media and utilize Natural Language Processing (NLP) and Large Language Models (LLMs) to synthesize structured documentation and adaptive learning pathways. By anchoring Retrieval-Augmented Generation (RAG) pipeline into the curriculum, CourseCraft AI ensures that student inquiries are answered strictly within the boundaries of the course content, maintaining factual fidelity and preventing model hallucinations.

## **2. Literature Review**

### **2.1. AI-Powered Personalized Learning Pathways**

The educational landscape is undergoing a profound transformation as Artificial Intelligence facilitates a transition from rigid, static curricula to bespoke learning experiences tailored to the idiosyncratic needs of individual students. Empirical evaluations of AI-driven adaptive platforms have demonstrated significant quantitative enhancements; for instance, students utilizing personalized instructional pathways exhibited a 25% improvement in grade attainment, test scores, and overall engagement compared to traditional pedagogical methods. To maximize instructional efficiency, these systems dynamically monitor cognitive load and modulate content complexity in real-time based on the learner's evolving state of knowledge.

### **2.2. Retrieval-Augmented Generation (RAG) in Video Education**

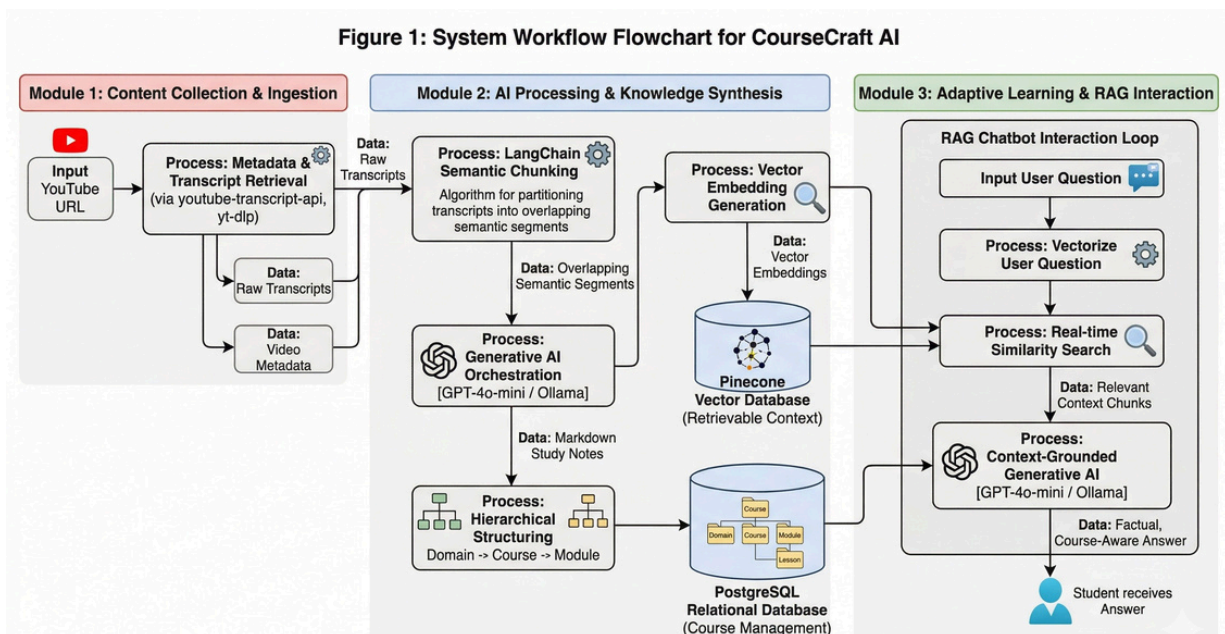
While Large Language Models exhibit significant capabilities, they frequently lack the specific context required to interrogate long-form instructional video media. Contemporary frameworks, such as VideoRAG, address this limitation by implementing multi-modal knowledge indexing and knowledge-grounded information retrieval. Rather than relying on generic text corpora, these advanced RAG systems transcribe lecture videos into structured textual data—utilizing sophisticated models like OpenAI's Whisper—and leverage vector databases to retrieve precise transcript segments. This methodology enables the generative module to synthesize context-aware responses, significantly reducing cognitive overload and enhancing learner satisfaction.

## 2.3. Educational Knowledge Graphs (EduKGs)

The synthesis of raw text into a hierarchical pedagogical structure necessitates advanced semantic modeling. Knowledge Graphs (KGs) have emerged as indispensable instruments for curriculum architecture and conceptual mapping within the educational domain. Research indicates that utilizing LLMs to extract core concepts and generate Hierarchical Tree-structured Knowledge Graphs allows platforms to automatically map domain relationships—spanning Domain, Course, Module, and Lesson tiers—directly from unstructured educational content.

## 2.4. Adaptive Multi-Agent Systems in E-Learning

To manage the inherent complexity of personalized instruction, current literature advocates for the implementation of multi-agent architectures over conventional rule-based engines. Within these environments, specialized AI agents—including automated profilers, content indexers, and recommendation modules—collaborate through reinforcement learning to dynamically orchestrate the sequence of learning objects. This collective intelligence ensures that the instructional pathway remains precisely aligned with the student's unique learning style and academic progress.



### 3. Methods

#### 3.1. System Architecture and Technology Stack

The technical realization of CourseCraft AI necessitates a highly resilient and scalable architectural framework, systematically partitioned into discrete functional layers to optimize operational efficiency and development velocity.

**Table 1: Software Requirements and Technology Stack**

| Layer       | Technologies                                                |
|-------------|-------------------------------------------------------------|
| Frontend    | Next.js, TypeScript, Tailwind CSS, shadcn/ui, Zod           |
| API Server  | Node.js, NestJS, TypeScript                                 |
| Database    | PostgreSQL, Prisma ORM, Pinecone DB (Vector DB)             |
| AI Pipeline | LangChain, OpenAI / Ollama, youtube-transcript-api, Whisper |
| Cloud       | AWS S3, Cloudflare CDN, AWS EC2 / Railway                   |

### 3.2. Organizational Distribution of Engineering Responsibilities

The delegation of technical labor is strategically bifurcated to maximize operational efficiency and accelerate development velocity:

**Table 2: Delineation of Proposed Work and Member Responsibilities**

| Project Member | Functional Domain                              | Engineering Responsibilities                                                                                                                                                                                                                                                                          |
|----------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shibani        | Frontend Development & System Integration      | Conceptualizing and constructing the user-centric interface utilizing the Next.js framework and Tailwind CSS. Developing comprehensive pedagogical views for course discovery, domain navigation, and media-grounded documentation while maintaining responsive design fidelity.                      |
| Subhankar      | Backend Infrastructure & AI Pipeline Synthesis | Engineering robust RESTful APIs through the NestJS framework and architecting the underlying relational database schema. Implementing the comprehensive AI pipeline for media processing, transcript extraction, and generative LLM orchestration, while managing cloud-native deployment strategies. |

### 3.3. Systematic Operational Workflow and Data Processing Pipeline

The systematic orchestration of the CourseCraft AI environment automates the conversion of unstructured audiovisual media into architecturally coherent pedagogical modules through the following sequential phases:

- 1. Content Ingestion and Metadata Acquisition:** The system programmatically retrieves pertinent metadata and textual transcripts from specified YouTube URLs, utilizing the *youtube-transcript-api* and *yt-dlp* utilities to ensure high-fidelity data capture.
- 2. Textual Preprocessing and Semantic Chunking:** To circumvent the inherent context window limitations of Large Language Models, extensive transcripts are algorithmically partitioned into overlapping semantic segments using the LangChain framework, thereby preserving contextual continuity.
- 3. Vector Embedding Generation:** The processed transcript segments are transformed into high-dimensional vector representations and subsequently persisted within the Pinecone Vector Database to facilitate efficient semantic retrieval.
- 4. Generative AI Orchestration:** The vectorized segments are synthesized through GPT-4o-mini or localized Ollama models to generate comprehensive, structured Markdown documentation that serves as the primary instructional material.
- 5. Hierarchical Knowledge Structuring:** Generated documentation is systematically organized into a strictly defined pedagogical hierarchy spanning the Domain, Course, Module, and Lesson tiers and archived within a PostgreSQL relational database for structured management.

**6. Context-Grounded Retrieval-Augmented Generation (RAG):** The system integrates an intelligent chatbot that vectorizes user inquiries to perform real-time similarity searches against Pinecone. This ensures that synthesized responses are grounded strictly within the retrieved course context, maintaining factual integrity.

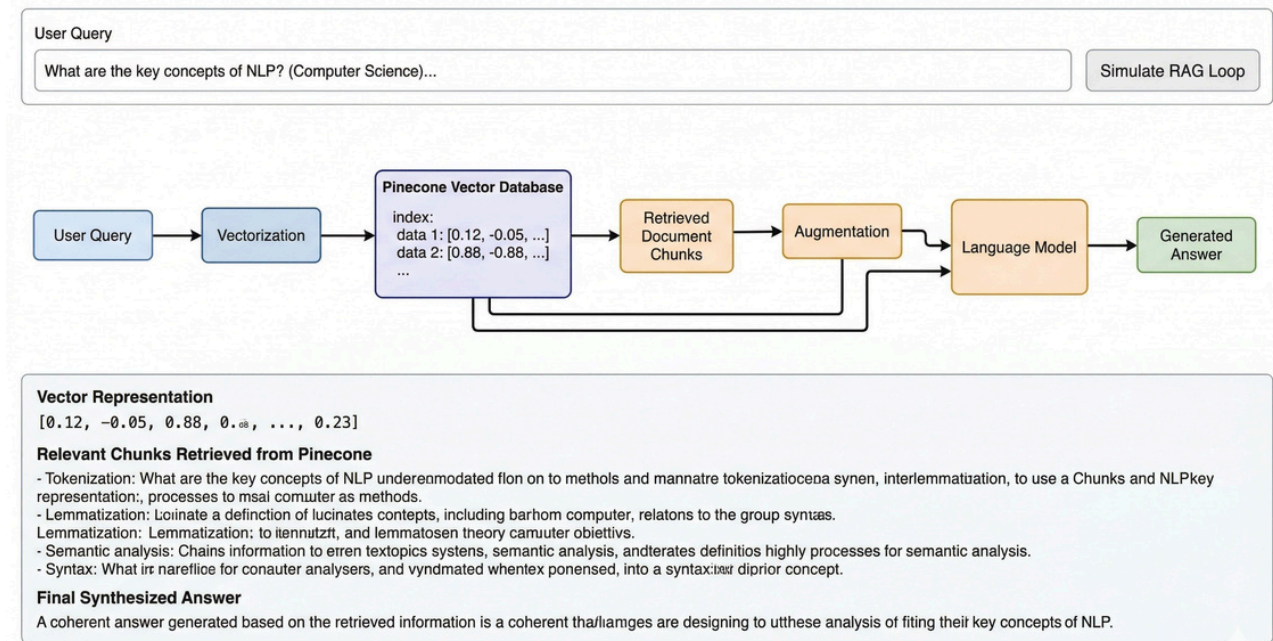
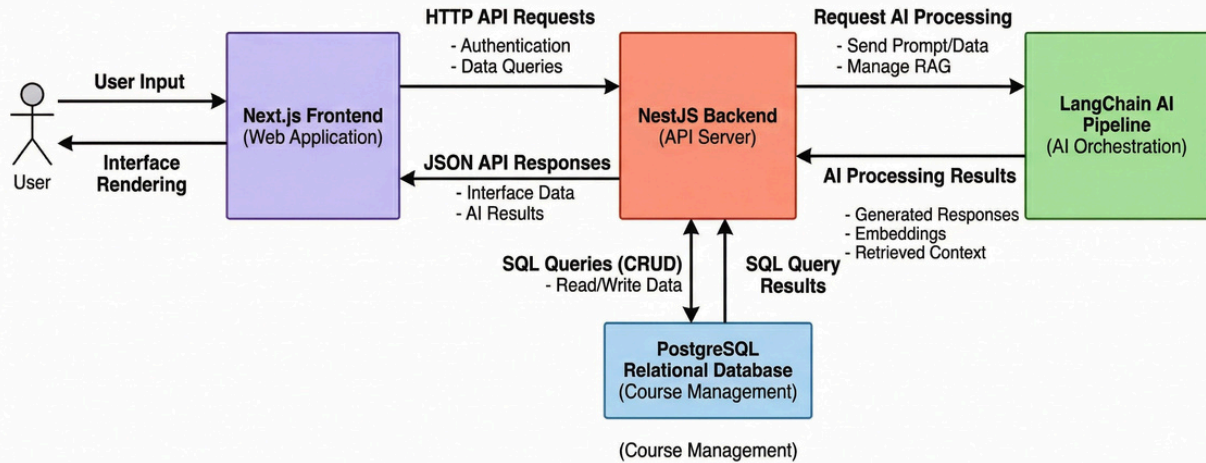
## 4. Results

To guarantee that synthesized curricula adhere to the most stringent paradigms of educational integrity, the system’s generative outputs are systematically appraised through rigorous, quantitative evaluation metrics.

**Table 3: Quality Assurance and Quantitative Evaluation Metrics**

| <b>Quantitative Evaluation Metric</b> | <b>Functional Application within the CourseCraft AI Ecosystem</b>                                                                                                                                                                  |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Faithfulness Score</b>             | Functions as the foundational safeguard against Large Language Model hallucinations, ensuring that the Retrieval-Augmented Generation module maintains strict factual fidelity to the source audiovisual media.                    |
| <b>Cognitive Load Index</b>           | Leveraged to audit the complexity of synthesized pedagogical documentation; where density indices exceed optimal thresholds, prompt engineering parameters are dynamically adjusted to facilitate clearer conceptual articulation. |
| <b>Relevance and Coherence</b>        | Employed to validate the semantic precision of the RAG chatbot, ensuring that user inquiries are resolved through contextually pertinent responses devoid of extraneous tangents.                                                  |

**Figure 2: System Architecture block diagram showing the interaction between the Next.js frontend, NestJS backend, PostgreSQL database, and the LangChain AI pipeline**



**Figure 3: Interactive RAG Loop for CourseCraft AI**

## **Expected Outcomes:**

1. A high-performance, responsive web application architecture capable of ingesting YouTube media links to generate architecturally coherent course hierarchies.
2. A sophisticated split-view learning interface that facilitates synchronized playback alongside AI-synthesized Markdown pedagogical documentation.
3. An intelligent, context-grounded RAG chatbot exhibiting minimal hallucination rates by strictly anchoring its generative processes to the embedded curriculum.

## **5. Conclusion and Future Works**

### **5.1. Conclusion**

The project effectively demonstrates the technical feasibility of **CourseCraft AI** leveraging advanced Natural Language Processing and generative AI architectures to mitigate the pervasive challenges associated with unstructured digital education. By facilitating a seamless transition from raw, open-source audiovisual media to structured, architecturally coherent learning environments, the platform substantially diminishes the cognitive load imposed on novice learners. The integration of algorithmic transcript chunking, high-dimensional vector embeddings via the Pinecone Vector Database, and context-grounded **Retrieval-Augmented Generation (RAG)** ensures that the pedagogical experience remains both personalized and anchored in factual integrity.

## 5.2. Future Works

While the current framework utilizes a structured rule-based engine for adaptive instructional pathing, future iterations of **CourseCraft AI** will transition toward a fully agentic AI architecture utilizing frameworks such as **LangGraph**. This evolution will enable specialized AI agents—including Strategist, Designer, and Coach agents—to collaborate through dynamic orchestration, simulating a personalized tutoring environment in real-time. Furthermore, migrating the backend infrastructure to self-hosted, quantized local Large Language Models via **Ollama** will optimize operational costs and enhance data privacy protocols.

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